Suyash Karnale
T012

Task (on house_price_regression_dataset.csv) (Don’t include in print document) Q.1 Find the correlation between Year_Built and House_Price.
Q.2 Create a correlation matrix for all numerical columns.
Q.3 Perform linear regression using:
X = Year_Built
Y = House_Price
Display the coefficient and intercept.
Q.4 Calculate the R2 score for the regression model:
Year_Built - House_Price
Q.5 Calculate the R2 score for:
Square_Footage - House_Price
Q.6 Calculate the R2 score for:
Neighborhood_Quality - House_Price
Q.7 Compare the R2 scores of all three regression models.
Code:-

```python
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = pd.read_csv("house_price_regression_dataset.csv")

print("Dataset:")
print(data.head())

data["Year_Built"] = 2026 - data["age"]
data["Square_Footage"] = data["house_size_sqft"]
data["Neighborhood_Quality"] = data["location_score"]
data["House_Price"] = data["house_price"]

correlation = data["Year_Built"].corr(data["House_Price"])

print("\nQ.1 Correlation between Year_Built and House_Price:")
print(correlation)

correlation_matrix = data.corr(numeric_only=True)

print("\nQ.2 Correlation Matrix:")
print(correlation_matrix)

X = data[["Year_Built"]]
Y = data["House_Price"]

model_year = LinearRegression()
model_year.fit(X, Y)

print("\nQ.3 Linear Regression: Year_Built - House_Price")
print("Coefficient:", model_year.coef_[0])
print("Intercept:", model_year.intercept_)
```

Sagar Kandari
T010

```python
Y_pred_year = model_year.predict(X)

r2_year = r2_score(Y, Y_pred_year)

print("\nQ.4 R2 Score for Year_Built - House_Price:")
print(r2_year)

X = data[["Square_Footage"]]

model_square = LinearRegression()
model_square.fit(X, Y)

Y_pred_square = model_square.predict(X)

r2_square = r2_score(Y, Y_pred_square)

print("\nQ.5 R2 Score for Square_Footage - House_Price:")
print(r2_square)

X = data[["Neighborhood_Quality"]]

model_quality = LinearRegression()
model_quality.fit(X, Y)

Y_pred_quality = model_quality.predict(X)

r2_quality = r2_score(Y, Y_pred_quality)

print("\nQ.6 R2 Score for Neighborhood_Quality - House_Price:")
print(r2_quality)

print("\nQ.7 Comparison of R2 Scores:")
print("Year_Built - House_Price:", r2_year)
print("Square_Footage - House_Price:", r2_square)
print("Neighborhood_Quality - House_Price:", r2_quality)

scores = {
    "Year_Built": r2_year,
    "Square_Footage": r2_square,
    "Neighborhood_Quality": r2_quality
}

best_model = max(scores, key=scores.get)

print("\nBest Predictor:", best_model)
print("Highest R2 Score:", scores[best_model])
```
✓ 2.9s

Suyash Karnale

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Output:-

```
Dataset:
   age  membership_years  number_of_children  house_size_sqft  bedrooms  \
0   58              12.7                   3             1841         1
1   48               9.9                   2             1259         5
2   34               6.8                   2             1187         3
3   62              11.7                   4             1596         3
4   27               5.2                   2             1912         1

   bathrooms  location_score  house_price
0          1               3       505286
1          3               4       573482
2          3               1       488645
3          1               1       465730
4          2               8       677596

Q.1 Correlation between Year_Built and House_Price:
0.03733773593654159

Q.2 Correlation Matrix:
                         age  membership_years  number_of_children  \
age                 1.000000          0.749623            0.568111
membership_years    0.749623          1.000000            0.429163
number_of_children  0.568111          0.429163            1.000000
house_size_sqft    -0.047545         -0.104603            0.158935
bedrooms            0.051522          0.076185            0.137076
...
Neighborhood_Quality - House_Price: 0.10034076936237868

Best Predictor: Square_Footage
Highest R2 Score: 0.673792388459706
```